

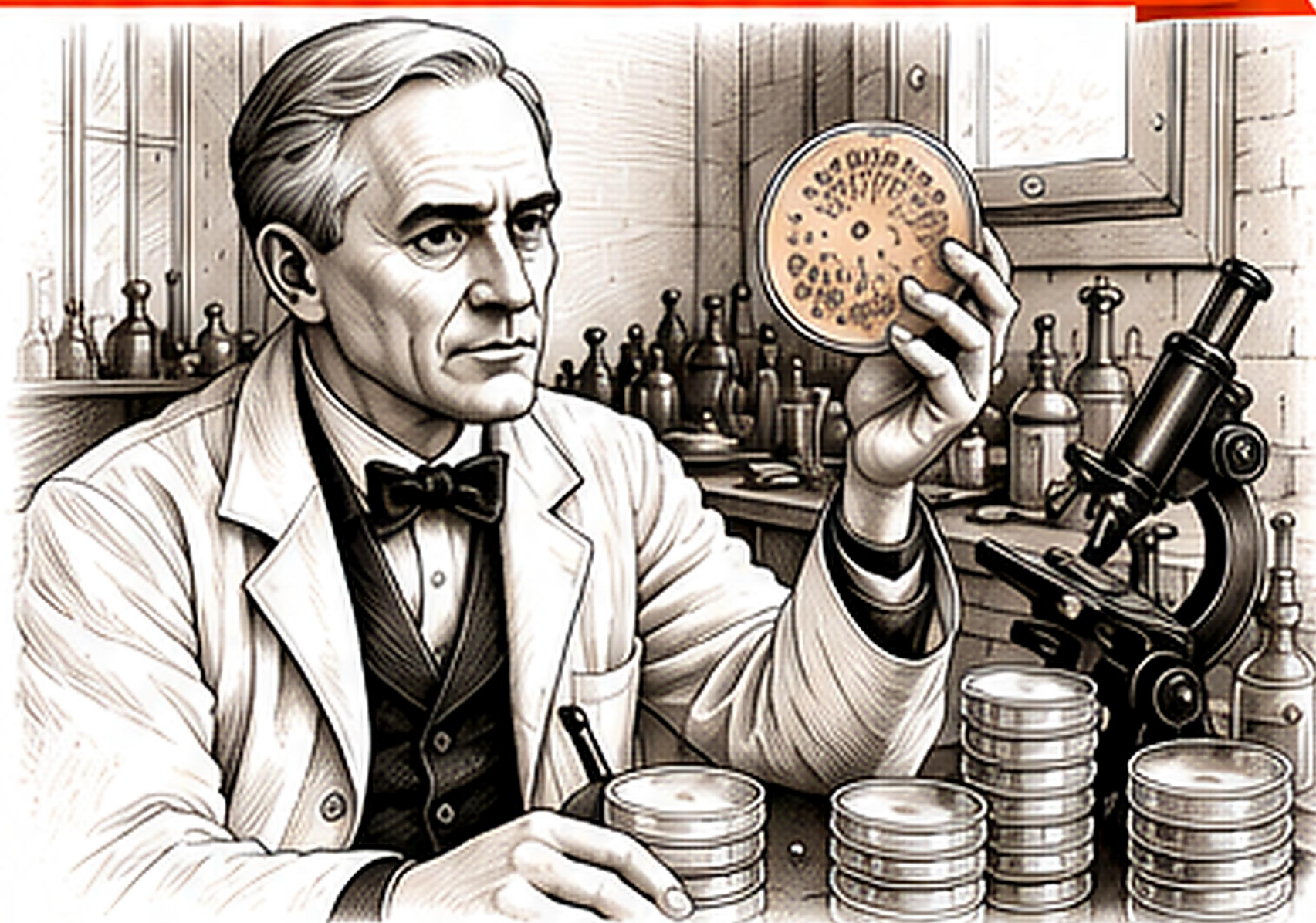
ALEXANDER FLEMING

SPARKWORKS

Think through anything.

THE MAN WHO CHANGED THE WORLD BY NOTICING

- ✓ It's 1928. Before antibiotics, infections killed millions.
- ✓ Doctors had no way to stop bacteria inside the body.
- ✓ Alexander Fleming was a Scottish doctor in London.
- ✓ He was messy. He wasn't famous. He was just curious.
- ✓ A dirty dish. A surprise. A question that changed the world.



EUREKA MOMENT

When something surprises you — you just learned something.

Fleming asked why.

HOW FLEMING FOUND PENICILLIN.

A messy lab. A moldy dish. A question no one else asked.

1 THE WORLD BEFORE ANTIBIOTICS



Cuts, sore throats, infections—they all could kill you.
Babies died. Soldiers died.
Millions died.
Doctors had no medicine to stop bacteria.

2 GROWING BACTERIA TO TEST



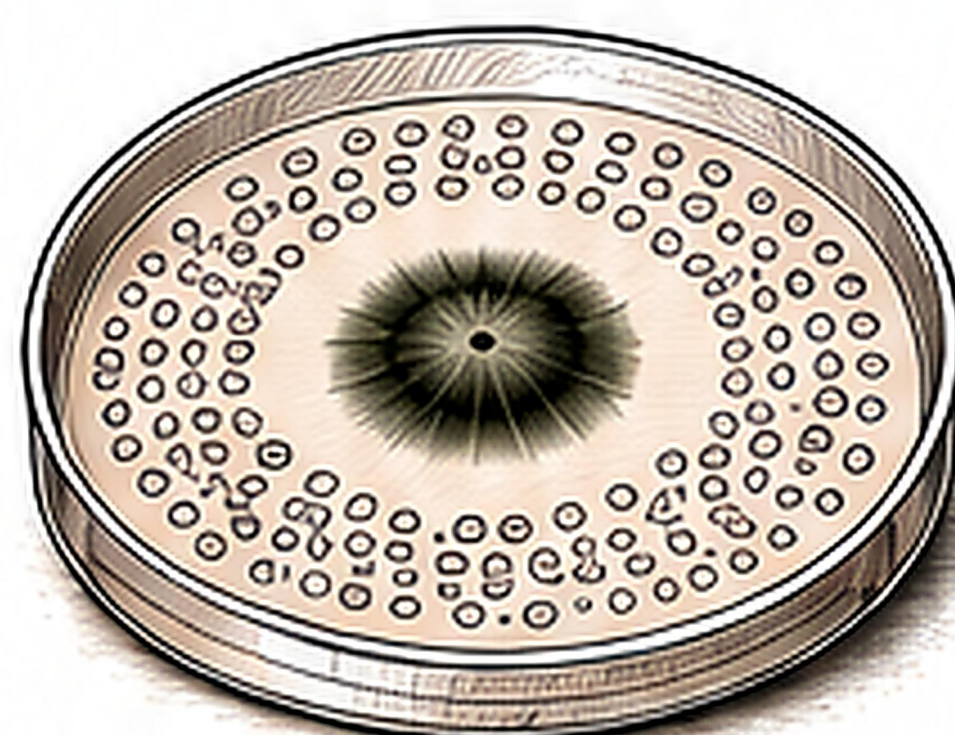
Scientists put a tiny drop of bacteria on jelly food in a dish.
They wait.
Millions grow into a visible colony.
Now you can test chemicals to see what kills them.

3 FLEMING'S MESSY LAB



Hundreds of dishes.
No magic medicine yet.
August 1928: Fleming goes on vacation for two weeks, leaving the dishes behind.

4 THE MOLD LANDS AND THE WEIRD RING



A mold spore lands on a dish and grows.
Sept. 3, 1928: Fleming comes back.
He sees a clear ring around the mold—no bacteria there. Only near the mold.
Far away, bacteria grow.
He stops. He asks why.

5 THE DISCOVERY PENICILLIN

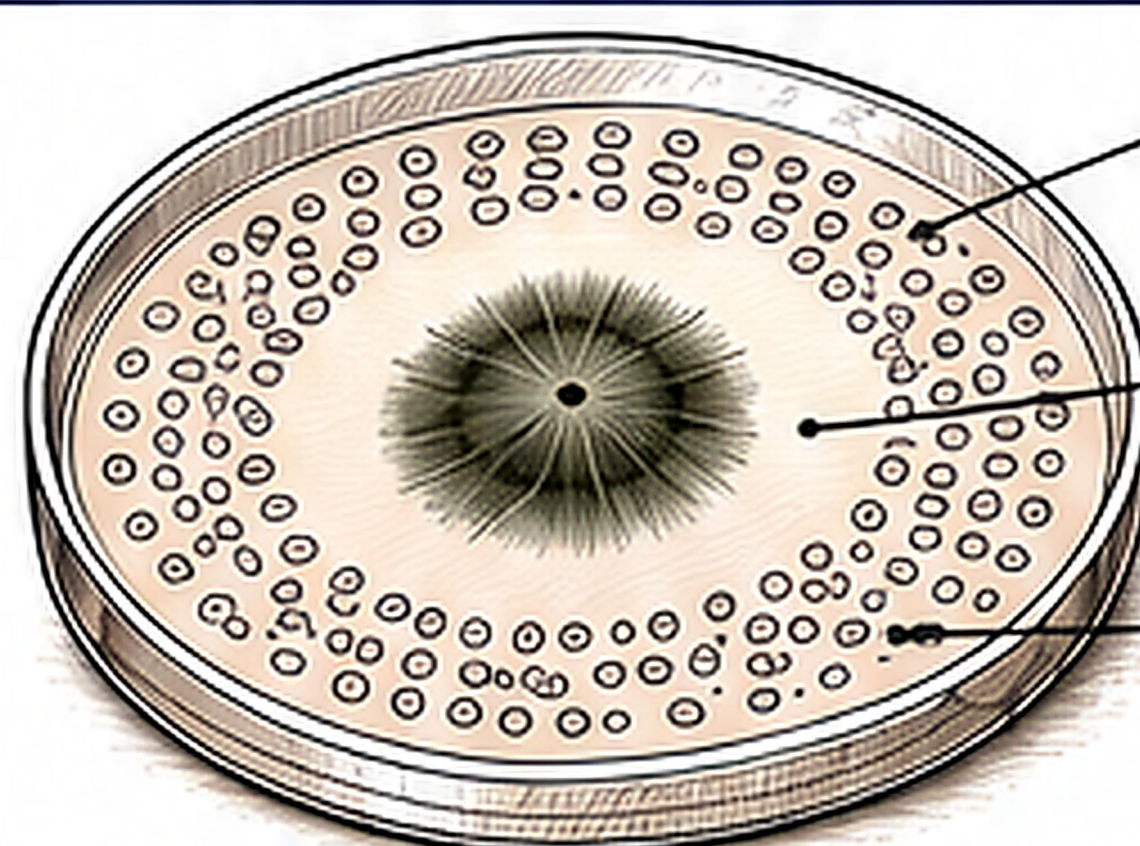


The mold is making a chemical that kills bacteria.
He names it penicillin.
Within 20 years, it saves soldiers. Within 50 years, it saves **200 million lives**.

SURPRISE = DATA

Why is the bacteria ONLY dead near the mold?

Something from the mold spreads out into the jelly.
Strong near the mold — it kills bacteria.
Weaker far away — bacteria survive.
A natural experiment Fleming almost threw away.



MOLD

The mold grows in the center.

CLEAR RING

No bacteria here. They are dead.

BACTERIA

Far away, they grow normally.

THE LESSON: SURPRISE IS AN OPPORTUNITY.

THE LESSON

NOTICE THE SURPRISE



Something unexpected happens.

DON'T IGNORE IT



Stop. Look closer. Ask why.

INVESTIGATE



Treat it like important data.

DISCOVER



You might find something powerful inside.

CHANGE THE WORLD



Big discoveries start with small moments.



THE REAL IMPACT

Penicillin turned infections from often-fatal into often-curable.
It launched the antibiotic era and has saved an estimated **200 MILLION LIVES**.



WWII

More soldiers died from infections than from battle.



1940s

Penicillin is mass-produced and used in WWII.



Lives saved.

Infections that once killed millions are now treatable.



A better world.

More people live. Healthier lives. Stronger communities.



OBSERVE
Look closely. Pay attention. Notice details.



QUESTION
Ask "why" or "what if" when things look weird.



TEST & EXPLORE
Gather evidence. Run experiments. Stay curious.



LEARN & SHARE
Turn discoveries into knowledge that helps others.



MAKE AN IMPACT
Use what you learn to solve problems that matter.

GREAT DISCOVERIES DON'T COME FROM PERFECT PLANS — THEY COME FROM **PAYING ATTENTION**.